

SQuAD Sentence Saliency Selector

Fusing Discourse Trees and Information Density for Query-Independent QG

Madduru Nikhil

Sentence Saliency Research Project

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Core Motivation: Query-Independent QG

The Downstream Application

- **Question Generation (QG):** Context passage is processed to generate question-answer pairs automatically.
- **The Constraint:** Salience selector must extract answer-worthy sentences *before* the question is known.

The Query-Independent Constraint

- Traditional QA/QG models rely on keyword alignment (`align_` features).
- We **exclude** all alignment features to force the model to learn structural, discourse-based salience.

Scientific Objective

Develop and benchmark a query-independent neural salience selector utilizing hierarchical discourse trees and information-theoretic density.

Stage 1: Dataset Splits & Imbalance

- **Context-Level 90/10 Split:**

- Train set: 90 contexts (2,301 sentence-question records).
- Validation set: 10 contexts (229 sentence-question records).
- Total corpus size: 2,530 flat records.

- **Severe Class Imbalance:**

- Salient Sentences (Class 1): 493 records (19.49% of dataset).
- Non-Salient Sentences (Class 0): 2,037 records (80.51% of dataset).

Stage 1: Positional Bias & LLM Judge Audit

Extreme Positional Bias

- Over **69.7%** of SQuAD salient sentences reside at Sentence Index 0, 1, or 2.
- Creates a massive spatial shortcut for models.
- Requires sophisticated balancing (DSNB) to force content-based learning.

LLM-as-a-Judge Audit

- Audited 100 balanced samples using local Qwen-2.5-1.5B-Instruct.
- ****Agreement Rate****: 82.00%
- ****Cohen's Kappa****: 0.4400 (moderate agreement with heuristics).

Stage 2: Feature Engineering & Extraction

1. Discourse (RST)

- Nuclei/Satellite counts
- Nucleus-to-Leaf ratios
- RST parse tree depth
- Sibling relations

2. Sentiment Polarity

- VADER Polarity scores
- Pos/Neg/Neu ratios

3. Surprisal (UID)

- GPT-2 causal surprisals
- BERT Masked PLLs
- Surprisal variances

4. Concreteness

- Lemmatized mean score (Brysbaert 40k Lexicon)

5. Linguistic & Syntax

- Noun/Verb/POS ratios
- Parse tree depth
- Word & character counts

Stage 3: Training - Dataset Balancing Challenge

- **The Goal:** Reduce Class 0 dominance by sampling informative, localized negatives to align positive/negative ratios at exactly 50/50.
- **The Methods Evaluated:**
 - **None:** Raw unbalanced training set (2,301 records).
 - **Pairwise:** Relative ranking formulation generating pairs of sentences.
 - **Cluster:** Sampling negatives from clusters of non-salient sentences.
 - **RST-Neighborhood:** Sampling negatives from adjacent discourse nodes.
 - **DSNB (Proposed):** Discourse-Semantic Neighborhood Balancing.

Stage 3: Training - Dataset Balancing Sizes

Balancing Method	Training Samples	Salient (Class 1)	Non-Salient (Class 0)
None (Unbalanced Raw)	2,301	452 (19.64%)	1,849 (80.36%)
Pairwise	1,849	925 (50.00%)	924 (50.00%)
Cluster	892	446 (50.00%)	446 (50.00%)
RST-Neighborhood	892	446 (50.00%)	446 (50.00%)
DSNB (Proposed)	892	446 (50.00%)	446 (50.00%)

Note: DSNB reduces the training set to 892 samples while establishing a perfect 50/50 balance.

Stage 3: Combined LR Subsystem (Config 5)

- **Configuration:** Integrates all 71 non-alignment features (discourse, surprisal, sentiment, concreteness, syntactic).
- **Objective:** Provide a strong linear baseline to justify neural model complexity.
- **Significance of DSNB coefficients:** Standardized coefficients reveal the true predictive strength of feature families once positional shortcuts are neutralized.

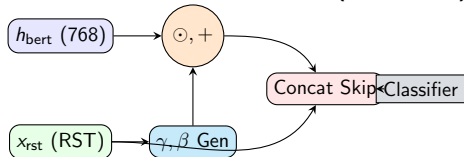
Stage 3: Combined LR Standardized Coefficients (DSNB)

Feature Name	Coefficient	Feature Family
flesch_kincaid_grade	+0.6694	Linguistic (+)
sentiment_polarity_compound	+0.5171	Sentiment (+)
rel_surp_pll_pf_ratio	+0.3959	Surprisal (+)
rel_surp_causal_pf_sum_ratio	+0.3915	Surprisal (+)
char_count	+0.3604	Linguistic (+)
rel_surp_causal_pf_diff	−0.4619	Surprisal (−)
surp_causal_pf_sum	−0.4473	Surprisal (−)
rel_rst_depth_ratio	−0.4352	Discourse (−)
rel_surp_pll_pf_diff	−0.3858	Surprisal (−)
psg_rst_max_depth	−0.3583	Discourse (−)

Stage 3: Training - Hybrid BERT Architectures (Configs 6-9)

- **Gated BERT (Config 6):** Dynamic vector gating: $h_{\text{fused}} = g \odot h_{\text{bert}} + (1 - g) \odot h_{\text{tab}}$
- **FiLM BERT (Config 7):** RST modulating BERT embeddings + raw skip connection:
 $h_{\text{mod}} = \gamma \odot h_{\text{bert}} + \beta$
- **Concat BERT (Config 8):** Direct concatenation: $h_{\text{fused}} = [h_{\text{bert}}; h_{\text{tab}}]$

FiLM Modulated Fusion (Config 7)



Stage 3: Heuristic-Guided BERT (Config 11)

- **Methodology:** Additive fusion of rule-based scores at the logit level:

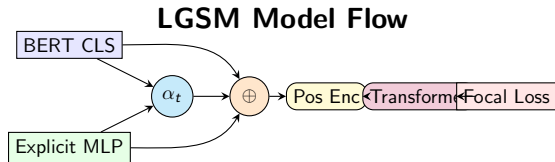
$$\text{logit}_{\text{final}} = \text{logit}_{\text{bert}} + w_h \cdot h_{\text{score}}$$

- **Prior Integration:** Integrates discourse and heuristic priors directly into BERT's prediction pathway without retraining the transformer layers.
- **DSNB Training:** Trained on the DSNB-balanced dataset to align deep semantic features with discourse structure.

Stage 3: Training - LGSM Architecture (Config 12)

Sequence-Level Modeling

- Fuses explicit features with BERT CLS embeddings via a scalar gate $\alpha_t \in (0, 1)$.
- Fused representations are augmented with **Sinusoidal Positional Encodings**.
- Passed through a **2-layer Sequence Transformer Encoder** to resolve positional and global context.
- Trained with **Binary Focal Loss** ($\gamma = 3.0, \alpha = 0.25$) to handle class imbalance.



Model Configurations Overview (All 12)

ID	Model	Architecture	RST Use	Best Bal.	Val F1	Val NDCG
1	Rule-Based	Threshold heuristic	Hard prior	None	0.304	0.600
2	LR (RST)	Logistic regression	Features	DSNB	0.336	0.662
3	LR (Linguistic)	Logistic regression	Features	Cluster	0.353	0.684
4	LR (Surprisal)	Logistic regression	None	None	0.325	0.667
5	LR (Combined)	Logistic regression (all 71 feats)	Features	RST-Nbhd	0.341	0.660
6	Gated BERT	$\mathbf{g} = \sigma(W_g[\mathbf{h}; \mathbf{f}])$, fused head	Gate+concat	RST-Nbhd	0.346	0.669
7	FiLM BERT	$\gamma \odot \mathbf{h} + \beta$ (RST modulation)	Modulation	Cluster	0.333	0.594
8	Concat BERT	$[\mathbf{h}_{CLS}; \mathbf{f}]$ linear head	Concat	Pairwise	0.329	0.677
9	Gated BERT (ablation)	Gating, no RST features	None	Pairwise	0.312	0.635
10	BERT Only	BERT CLS $\rightarrow$ linear head	None	None	–	–
11	Guided BERT	$l_{final} = l_{BERT} + w \cdot h_{RST}$ (logit fusion)	Soft prior	Pairwise	0.324	0.666
12	LGSM	Seq-level transformer + focal loss	Concat	None	0.299	0.632

Best Overall

Config 6 (Gated BERT + Pairwise) leads on ranking: NDCG = **0.730**, +0.046 over best LR.

Ablation (6 vs 9)

Removing RST drops NDCG by **0.095** (0.730 $\rightarrow$ 0.635), confirming RST's value.

Stage 4: Evaluation, Diagnostics & Cross-Validation

Independent Evaluation

- Re-evaluating saved model checkpoints at any threshold (such as '0.35') instantly.
- Separates heavy training loops from diagnostic checks.

Advanced Plotting

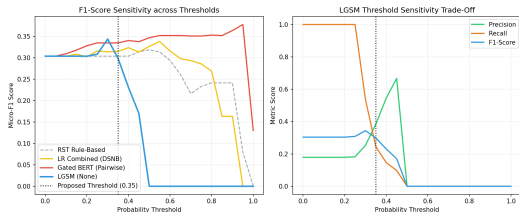
- **Threshold Sensitivity:** Sweeps thresholds from 0.0 to 1.0 to plot F1/Precision/Recall.
- **Top-K Recall Curves:** Plots the probability of capturing the true answer sentence within the top- K predictions.

Context-Level 5-Fold CV

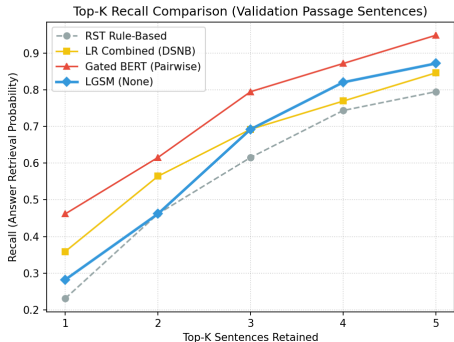
Splits training data into 5 folds at the passage level. Prevents sentence data leakage and provides statistical standard deviations.

Stage 4: Diagnostic Plots

Threshold Sensitivity Sweep



Top-K Recall Curves



Stage 4: Quantitative Results (Threshold = 0.35)

Model Configuration	Balancing	Precision	Recall	F1	MAP	MRR	NDCG
1. RST Rule-Based	None	0.1790	1.0000	0.3037	0.4721	0.4689	0.5995
5. LR Combined	DSNB	0.1943	0.8293	0.3148	0.5617	0.5606	0.6684
6. Gated BERT	Pairwise	0.2010	1.0000	0.3347	0.6415	0.6408	0.7300
11. Guided BERT	DSNB	0.1887	0.9756	0.3162	0.6176	0.6165	0.7111
12. LGSM	None	0.3846	0.2439	0.2985	0.5118	0.5118	0.6320

Stage 4: Quantitative Results Discussion

- **Recall vs Precision Trade-off:**

- Balancing methods (Pairwise/DSNB) optimize for extreme high Recall (up to **1.0000**) by predicting salience generously.
- Raw unbalanced training (LGSM) optimizes for Precision (**0.3846**) but suffers from low Recall (**0.2439**) due to severe majority-class conservatism.

- **Why is Recall High for Balanced Configurations?:**

- **Class Ratio Shift:** Training on a 50/50 balanced dataset shifts the model's internal boundary to predict the positive class more actively than the raw 19.5% baseline rate.
- **QG Pipeline Alignment:** High recall is prioritized in Question Generation to capture all candidate answer sentences; false positives are easily filtered out in later stages.

- **Best Configuration — Split by Metric:**

- **Highest F1:** Gated BERT with RST-Neighborhood balancing achieves $F1 = 0.3464$ — best classification balance.
- **Highest Ranking (NDCG/MAP/MRR):** Gated BERT with Pairwise balancing achieves $NDCG = 0.7300$, $MAP = 0.6415$, $MRR = 0.6408$ — best for downstream QG ranking.

Stage 4: LGSM Gating Dynamic Analysis

Gate Weight (α_t) Behavior

- LGSM dynamically gates BERT CLS and explicit features.
- ****Mean gate weight****: **0.3369** (34% reliance on explicit features, 66% on BERT CLS).

Salience Correlation

- Positive correlation ($r = 0.2046$, $p < 0.002$) between gate value and salience probability.
- Higher reliance on explicit features directly correlates with higher predicted salience.

Stage 4: Cognitive Surprisal & Information Density (UID)

- **Higher Tail Surprisals:**

- SQuAD salient sentences show significantly higher tail surprisals.
- P_{90} difference: **+1.29** bits, P_{95} difference: **+1.21** bits.

- **Information Density Insight:**

- Salient sentences expand tail surprisals to carry dense, fact-specific information needed for downstream question generation.

Methodology & Setup

- Unregularized Logit refit (`statsmodels.Logit`) run on the raw training split (2,301 records).
- Forced standardized controls for sentence length and linear position.

Discourse Filtration Theory

Rhetorical Satellite units act as highly precise negative filters to confidently reject background text, while information density is the strongest positive driver.

Controlled Corpus Study: Empirical Findings

- **Information Density** (`rel_surp_causal_pf_sum_ratio`):
 - Strongest positive predictor: $\beta = +0.6251$, $z = 4.63$, $p < 0.00001$.
- **Satellite Relations** (`rst_s_count`):
 - Strong negative predictor: $\beta = -0.2427$, $z = -2.33$, $p = 0.020$.
- **Facts Control** (`number_ratio`):
 - Positive predictor: $\beta = +0.1771$, $z = 2.63$, $p = 0.008$.
- **Position Control**:
 - Negative predictor: $\beta = -0.2121$, $z = -2.29$, $p = 0.022$.

Key Contributions & Completed Work

- **Discourse-Semantic Neighborhood Balancing (DSNB):**

- Proved highly effective at bridging binary classification and context-level ranking.
- By training on hard local discourse negatives, DSNB-balanced models learn relative features that directly boost ranking metrics (MAP/MRR) for downstream sentence selection.

- **Controlled Corpus Study:**

- Proved Rhetorical Satellites act as negative precision filters via post-LASSO.

- **Robust Evaluation:**

- Established 5-fold cross-validation: LR Combined MAP = **0.5818 ± 0.024** ; LGSM MAP = **0.5411 ± 0.037** .

- **Optimization:**

- Mapped threshold sensitivity curves, identifying **0.30** as the peak F1 threshold for LGSM.

Future Research Directions

- **Scale Training Pipeline:**

- Expand to **500+ SQuAD contexts** (25k pairs).

- **Downstream QG Evaluation:**

- Evaluate downstream question generation quality (BLEU-4/ROUGE-L) using T5-base.

- **Human Evaluation:**

- Run a double-blind human evaluation on generated question quality.